

Mine Rescue Rover · Chassis, Enclosures & Electronics Assembly

Tier 1 skeleton, Tier 2 gas deck, Tier 3 compute deck, roof, walls, doors, fasteners and component placement. The 4WD drivetrain under the base plate is a separate document (next part). Every placement in this document comes from one coordinate model that is checked by script for clashes, fit and bolt length (results on the last page: **0 errors**).

Corrections applied (vs the Gemini spec)

#	Gemini spec	This document	Why
1	M4x12 bolts	M4x20 everywhere structural ; stack: head · metal washer 0.8 mm · plate · bracket · 1.0 mm rubber washer · M4 nyloc (5.0 mm)	Your instruction; lengths checked per joint (sheet 3)
2	Walls inside the plates (end walls 29.6 / 21.6 cm)	Walls outside every plate, zero gap. Bracket outer faces flush with the plate edge; plate sizes unchanged	Your instruction; deck dimensions never change
3	Walls 9.0 cm	Walls cover the plate edges: Tier 1 = 9.7 (0.4 + 9.0 + 0.3), Tier 2/3 = 9.3 (9.0 + 0.3)	You chose "keep it big"
4	Pillars tapped 15 mm	Tapped 20 mm (drill 3.3 mm to 23 mm). M4x20 through 4 mm base engages 15.2 mm and bottomed out in a 15 mm thread	Bolt would not clamp
5	Walls "bolted into pillar side faces" (never tapped)	Drill 4.2 mm clean through the pillar , washer + nyloc inside (stack 18.6 mm, 1.4 mm past nut). Side-wall holes Z 3.0/6.0, end-face holes Z 3.5/6.5 so they never cross	An M4x20 cannot be threaded into a 10 mm pillar side
6	One wall per box bolted with external brackets only	Rear wall of each tier = hinged access door: piano hinge at rear-left, 2 M4x20 latch bolts at rear-right (Tier 1 into the RR pillar, Tier 2/3 into a 45 mm latch post). Rear brackets stay as door stops	Hand access for lip bolts; no nuts needed inside a closed box
7	Gas ports 8.4 x 3.1 cm for flat MQ boards	Ports 8.5 x 4.40 cm, open at the bottom: MQ modules stand vertical (32 mm PCB) and their domes face out. Open bottom lets the box drop over the jutting sensors	Vertical sensors are 4.30 cm tall
8	Thermal "flat", arches 2.5 cm, L/R arches 0.8 cm off the minis	MLX90640 110° stands on its header facing the wall (a flat board looks at the ceiling). Arches 4.9 x 3.5 cm; left/right minis moved to Y 13.2-17.9 so all 4 lenses sit on the arch centres	Unclipped 110°x70° view; 4 x 110° = full 360° coverage
9	Camera holes Z 5.75	Camera holes Z 6.80	A camera at 5.75 hangs into the thermal sensor's upper field of view
10	No Pi battery	Battery C bay beside the 12 V bus (X 2.3-9.5, Y 2.3-15.5) + 5.1 V regulator bay (X 21.0-27.0, Y 11.0-15.5)	Your instruction; Pi 5 + hub isolated from logic power
11	RPLIDAR slot 7.1 x 7.1	Real A1M8 envelope 7.03 x 9.68 x 5.5 at dead centre	Datasheet size

Sheet 1 · Components and exact sizes used

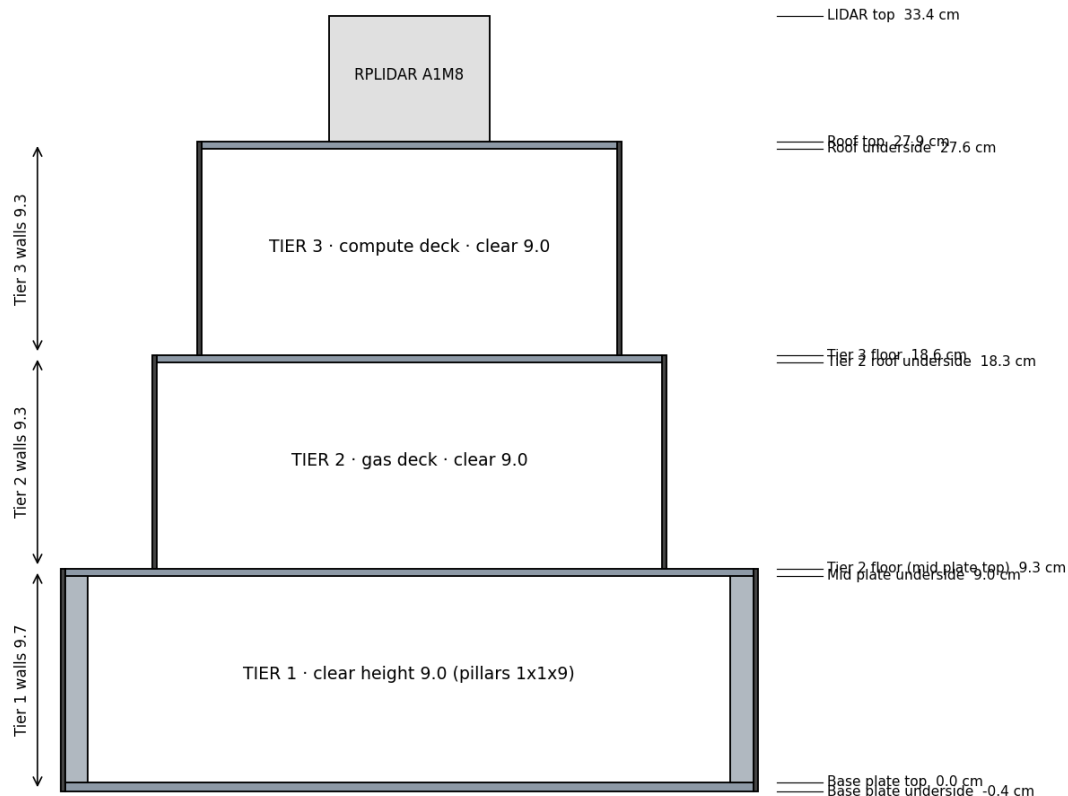
Item	Model chosen	Size used in the model (cm)	Notes
Gas sensors x6	MQ-2/4/7/8/135/136 FC-22 style modules	PCB 3.2 x 2.2, envelope 3.2 x 2.2 x 2.7, dome Ø2.0	Published module size 32x22x27 mm. Standing vertical on the 4-pin header
Thermal x4	Adafruit MLX90640 4469 (110° x 70°)	2.58 x 1.78 x 1.05	Wide version chosen for maximum coverage; 3.3-5 V supply, runs on ESP32 3V3
400-pt breadboards x5	standard	8.2 x 5.5 x 0.85	Fixed with their adhesive backing (no holes exist for standoffs)
170-pt minis x4	standard	4.7 x 3.5 x 0.85	
USB hub	Waveshare USB-HUB-7U	reserved bay 11.8 x 7.5 x 3.0	7 x USB 2.0, MTT, DC 5 V terminal input only (matches Battery C), metal case with mounting holes. Check its outline drawing (Waveshare wiki, image only) against the bay before buying.
Battery C (Pi + hub)	2S 7.4 V LiPo + 5.1 V step-down	bay 7.2 x 13.2 x 4.5; regulator bay 6.0 x 4.5 x 2.5	Pack: 2S, 3000-4000 mAh, >=20C, case <= 13.0 x 7.0 x 4.5. Regulator: 5.1 V out, >=8 A continuous (Pi 5 up to ~5 A + hub 1.5-2.5 A). Pick a pack by its datasheet case size, not the listing photo
Raspberry Pi 5	8 GB	8.5 x 5.6, 3.0 tall on 1.0 standoffs	M2.5 holes 58 x 49 mm, 3.5 mm from edges
RPLIDAR	Slamtec A1M8	9.68 x 7.03 x 5.5	Mount through its own base holes; match-drill the roof from the unit
DHT22	3-pin module	1.5 x 2.0 footprint, 2.51 tall	Roof front-left
Brackets	20 x 20 x 2 mm aluminium angle, 20 mm wide	2.0 x 2.0 x 2.0	One Ø4.2 hole per leg at 11 mm from the corner
Pillars / latch posts	10 x 10 mm solid aluminium	1 x 1 x 9.0 / 1 x 1 x 4.5	Tapped M4 20 mm deep at bolted ends
Hinge	aluminium piano hinge, 50 mm open (2 x 25 mm leaves)	cut to 8.0 (T1) / 4.4 (T2, T3) long	Knuckle on the outside rear-left corner

Sheet 2 · Plate and wall cut list

Part	Qty	Size (cm)	Material	Machining
Base plate (Tier 1 floor)	1	30.0 x 40.0	4.0 mm Al	12 lower-bracket holes, 4 pillar holes (under pillars, bolt up), driver/bus/battery fixings, drivetrain pattern (separate doc)
Mid plate (T1 roof / T2 floor)	1	30.0 x 40.0	3.0 mm Al	12 upper-bracket + 4 pillar holes, 8 Tier-2 lip holes, hinge-side clearance, H-T2 Ø16 at (15.0, 26.2), H-T3 Ø25 at (22.0, 25.7)
Tier 2 roof (T3 floor)	1	22.0 x 35.0	3.0 mm Al	8 roof-bracket holes, latch-post hole, 8 Tier-3 lip holes, H-T3 Ø25 at T2-local (18.0, 23.2)
Roof	1	18.1 x 31.1	3.0 mm Al	8 roof-bracket holes, latch-post hole, H-DHT22 Ø16 at (3.2, 4.2), H-LIDAR Ø16 at (9.05, 21.5), LIDAR holes match-drilled
T1 left / right side walls	2	40.2 x 9.7	2.0 mm Al	Front end overlaps front wall; rear end flush with plate edge (door covers it)
T1 front wall	1	30.0 x 9.7	2.0 mm Al	Sits between the side walls
T1 rear door	1	30.4 x 9.7	2.0 mm Al	Covers side-wall ends
T2 side walls	2	35.2 x 9.3	2.0 mm Al	MQ ports Y 26.6-35.0 x 0-4.40
T2 front wall	1	22.0 x 9.3	2.0 mm Al	MQ port X 6.75-15.25 x 0-4.40
T2 rear door	1	22.4 x 9.25	2.0 mm Al	Solid; 0.5 mm above the floor so it swings
T3 side walls	2	31.3 x 9.3	2.0 mm Al	Arch Y 13.1-18.0 x 0-3.5, camera hole (15.55, Z 6.8)
T3 front wall	1	18.1 x 9.3	2.0 mm Al	Arch X 6.6-11.5 x 0-3.5, camera hole (9.05, Z 6.8)
T3 rear door	1	18.5 x 9.25	2.0 mm Al	Arch + camera hole as front (shifted +0.2 by the overlap)
Pillars	4	1 x 1 x 9.0	10 mm Al bar	Ends tapped M4 x 20 deep; 2 through-holes per outer face (see #5)
Latch posts	2	1 x 1 x 4.5	10 mm Al bar	Top tapped 20 deep; door taps (Y) at Z 4.9 and 6.5; side through-hole (X) at Z 5.7
L-brackets	24 (T1) + 16 (T2) + 16 (T3)	20 x 20 x 20 mm	2.0 mm Al	T1 12 lower + 12 upper; T2/T3 8 lips + 8 roof brackets each (2 per wall)

Sheet 3 · Heights and fasteners

Front elevation · Z stack (cm, Z = 0 at base plate top) · walls outside every plate

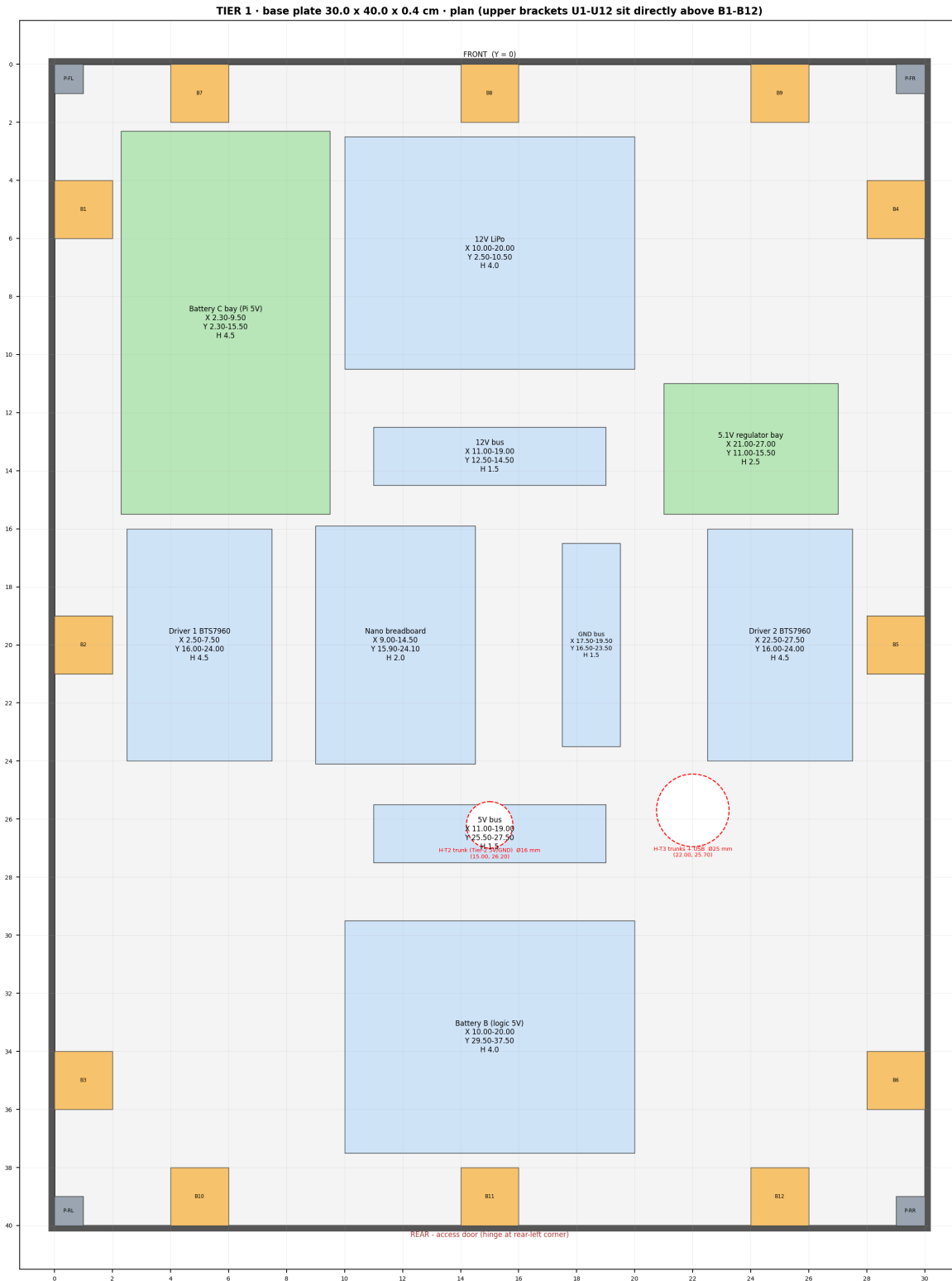


Drivetrain (under base plate) - separate document. Base plate underside is 14.0 cm above the floor.

Joint	Stack (mm)	Past the nyloc
Bracket to 3.0 mm plate	washer 0.8 + plate 3.0 + bracket 2.0 + rubber 1.0 + nyloc 5.0 = 11.8	8.2 mm
Bracket to 4.0 mm base plate	0.8 + 4.0 + 2.0 + 1.0 + 5.0 = 12.8	7.2 mm
Wall to bracket leg	0.8 + wall 2.0 + bracket 2.0 + 1.0 + 5.0 = 10.8	9.2 mm (points inward, all checked clear)
Wall to pillar / latch post (through)	0.8 + 2.0 + 10.0 + washer 0.8 + nyloc 5.0 = 18.6	1.4 mm (2 threads, enough for the nylon insert)
Plate into pillar end (tapped)	0.8 + 4.0 (base) or 3.0 (mid) -> 15.2 / 16.2 mm of thread	tap depth 20 mm, Loctite blue
Door into pillar / post (tapped)	0.8 + door 2.0 -> 17.2 mm, exits 7.2 mm inside the corner	corner area is clear
Hinge leaf to wall	0.8 + leaf 1.0 + wall 2.0 + 0.8 + 5.0 = 9.6	10.4 mm

Rubber washers go only where a bolt passes an L-bracket (spec 7.2). Every lip bolt tip that hangs into the deck below stops 1.58 cm under that plate; the script confirms nothing sits there.

Sheet 4 · Tier 1 skeleton (base plate)

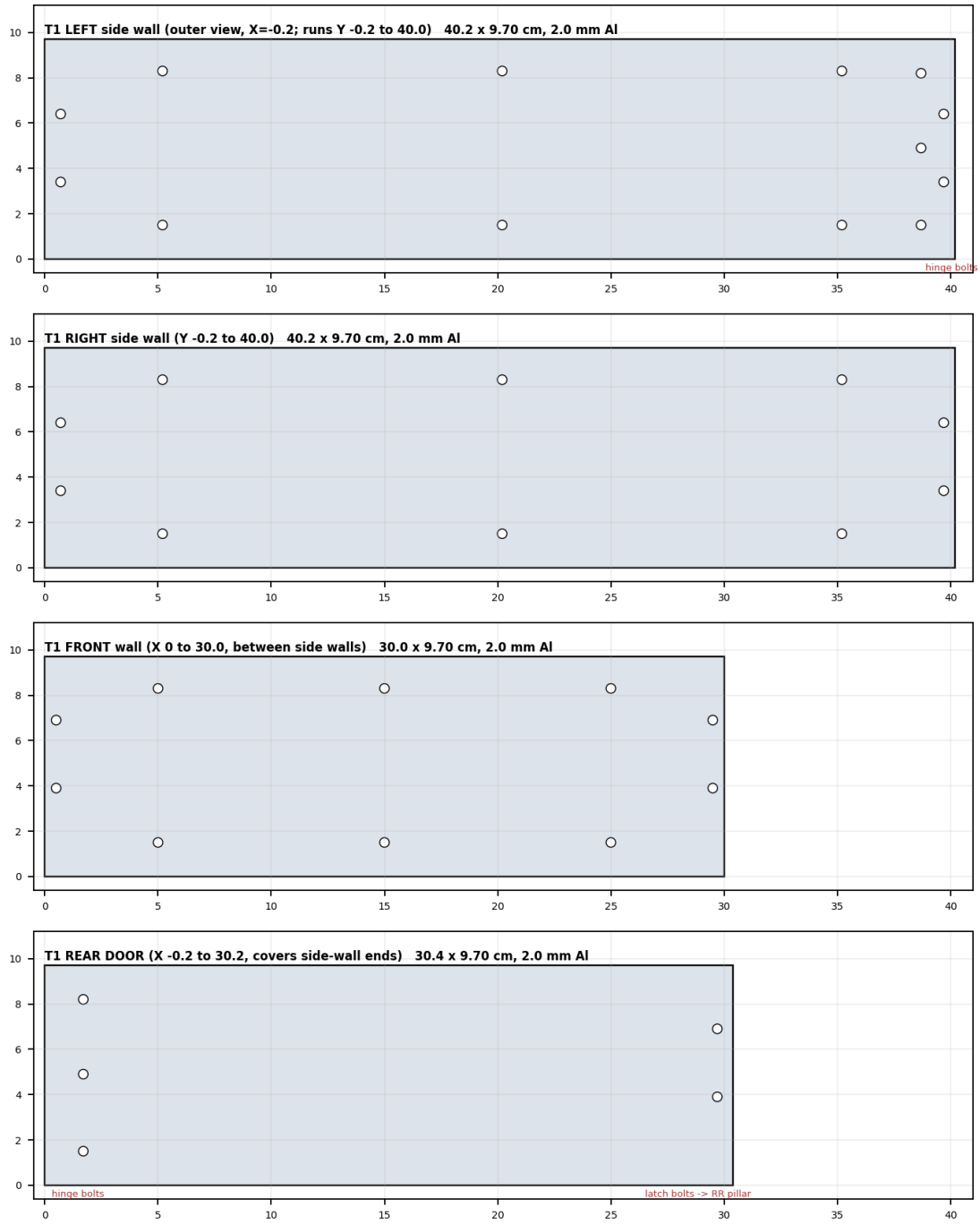


Dashed red circles are holes in the **mid plate above** (wire pass-throughs). Upper brackets U1-U12 are directly above B1-B12, bolted to the mid plate. B10-B12 / U10-U12 are door stops (not bolted to the door).

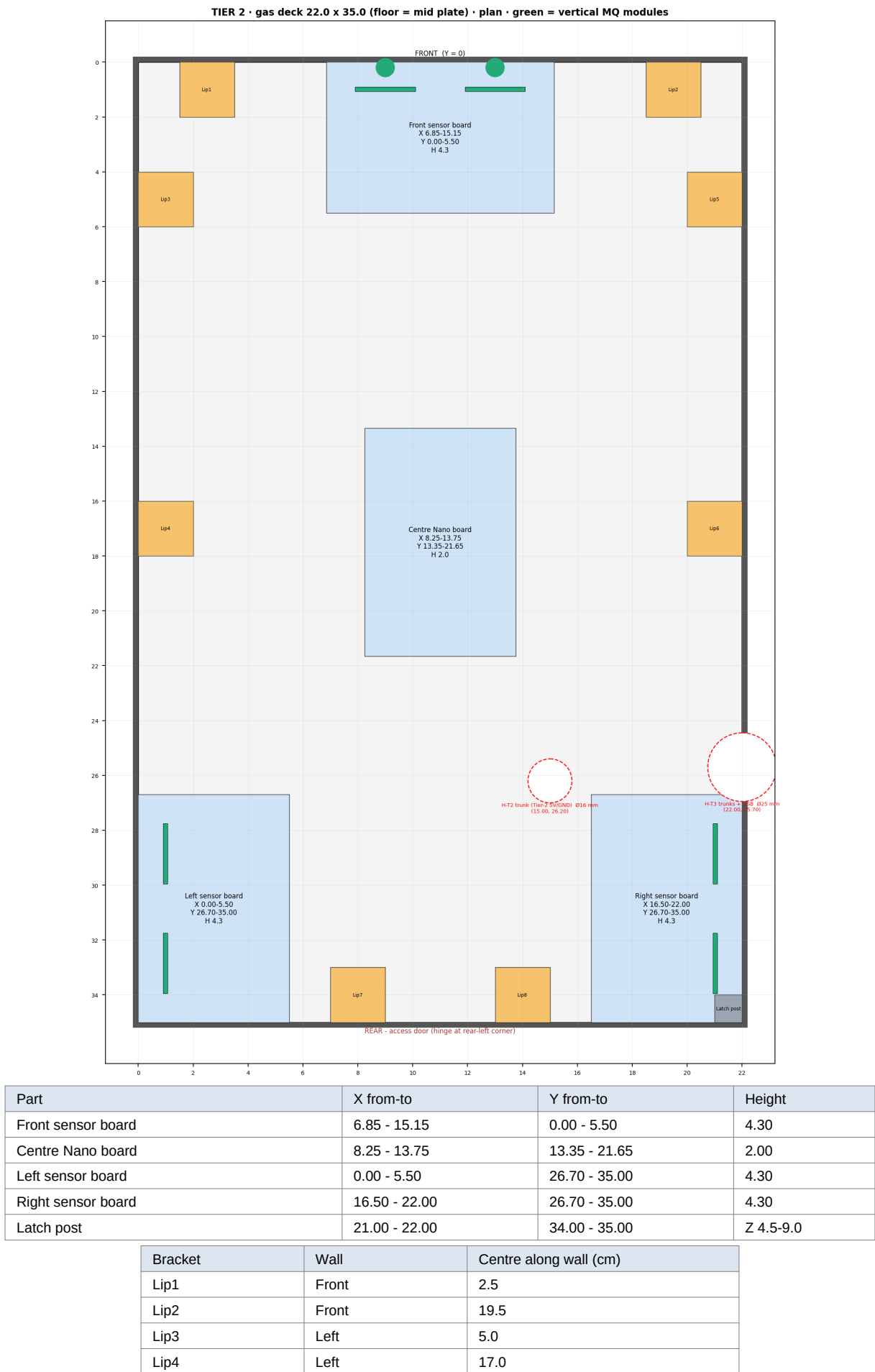
Part	X from-to	Y from-to	Height
P-FL	0.00 - 1.00	0.00 - 1.00	9.00
P-FR	29.00 - 30.00	0.00 - 1.00	9.00
P-RL	0.00 - 1.00	39.00 - 40.00	9.00
P-RR	29.00 - 30.00	39.00 - 40.00	9.00
12V LiPo	10.00 - 20.00	2.50 - 10.50	4.00
12V bus	11.00 - 19.00	12.50 - 14.50	1.50
Driver 1 BTS7960	2.50 - 7.50	16.00 - 24.00	4.50
Nano breadboard	9.00 - 14.50	15.90 - 24.10	2.00
GND bus	17.50 - 19.50	16.50 - 23.50	1.50
Driver 2 BTS7960	22.50 - 27.50	16.00 - 24.00	4.50
5V bus	11.00 - 19.00	25.50 - 27.50	1.50
Battery B (logic 5V)	10.00 - 20.00	29.50 - 37.50	4.00
Battery C bay (Pi 5V)	2.30 - 9.50	2.30 - 15.50	4.50
5.1V regulator bay	21.00 - 27.00	11.00 - 15.50	2.50

Bracket	Wall	Centre along wall (cm)
B1	Left	5.0
B2	Left	20.0
B3	Left	35.0
B4	Right	5.0
B5	Right	20.0
B6	Right	35.0
B7	Front	5.0
B8	Front	15.0
B9	Front	25.0
B10	Rear	5.0
B11	Rear	15.0
B12	Rear	25.0

TIER 1 walls · Z measured from wall bottom edge (= base plate underside) · all holes Ø4.2



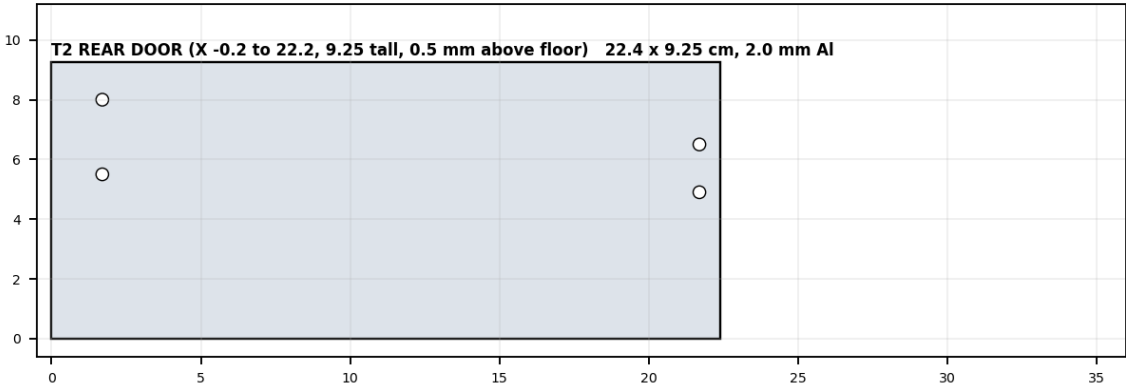
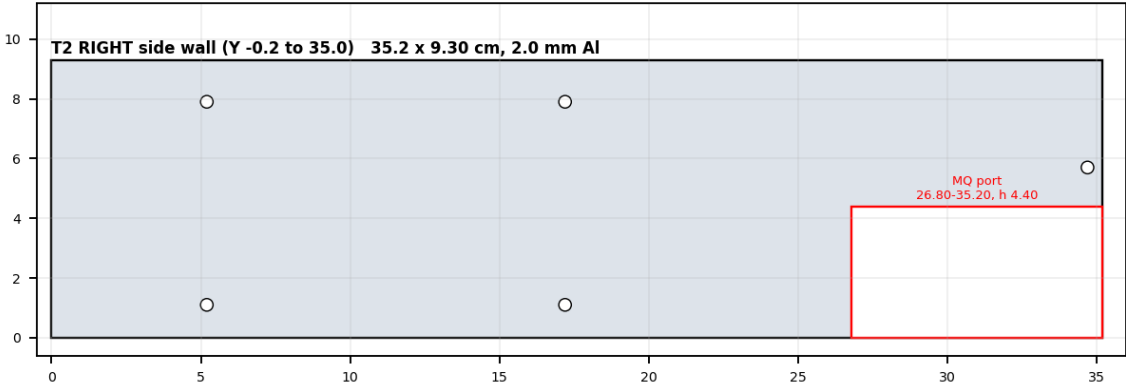
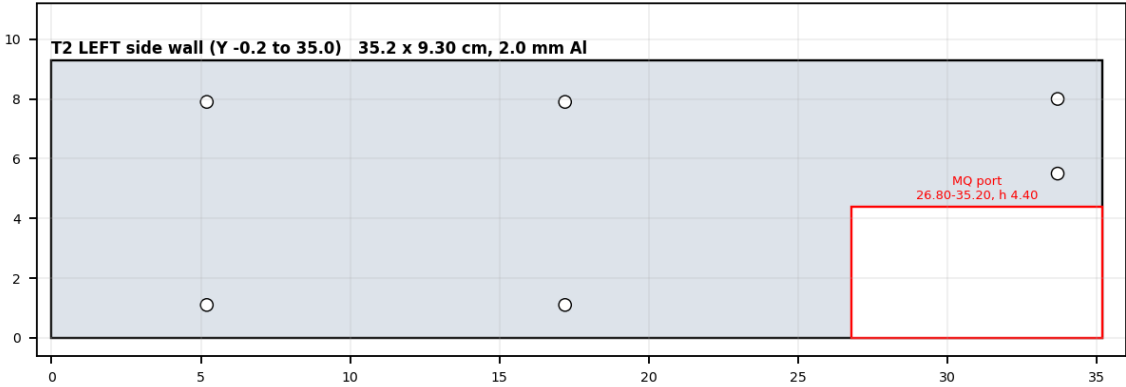
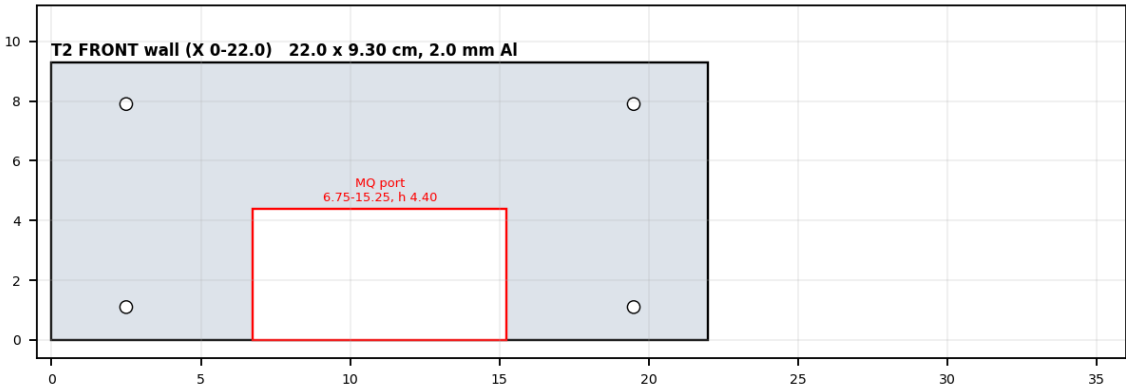
Sheet 5 · Tier 2 gas deck



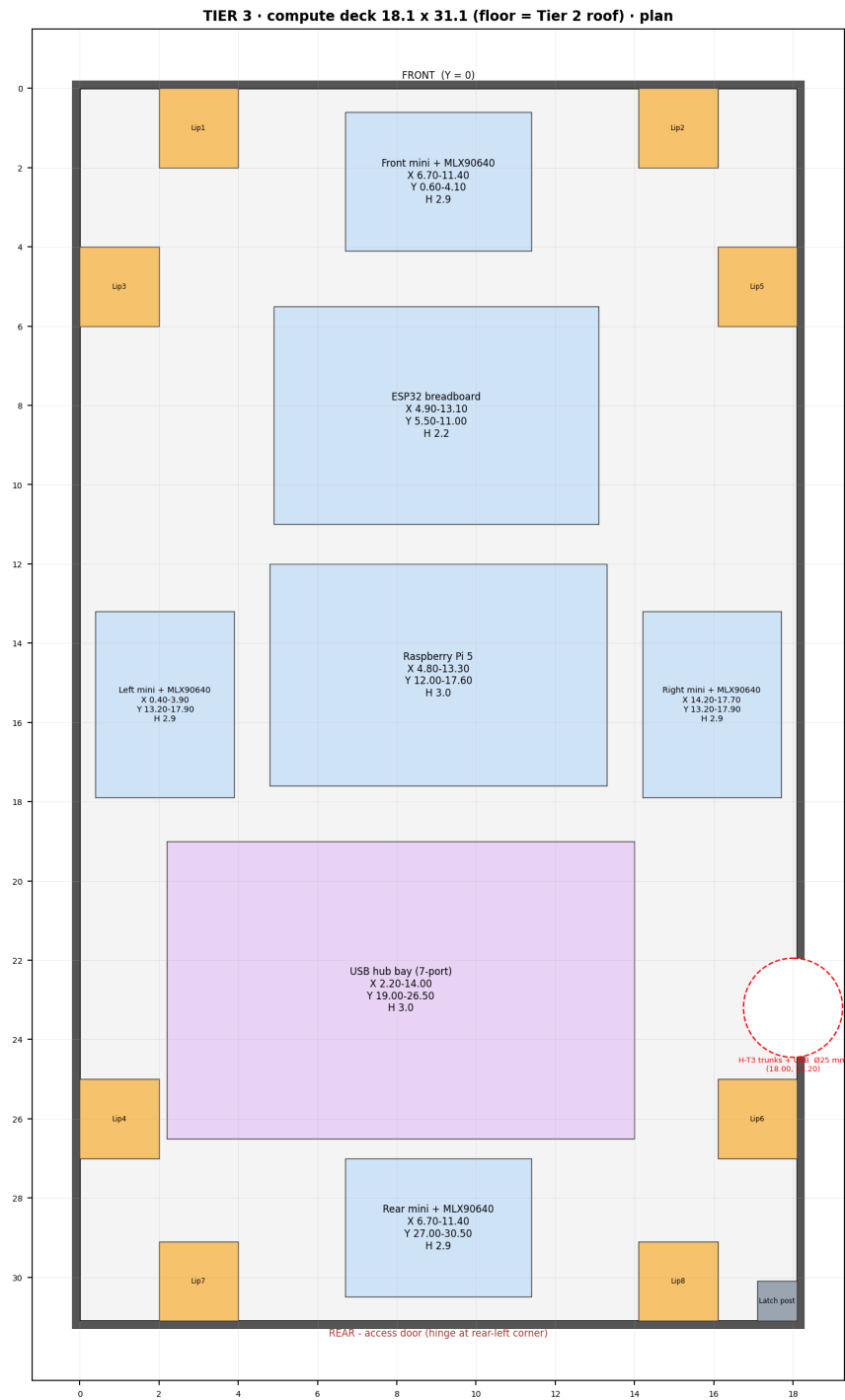
Bracket	Wall	Centre along wall (cm)
Lip5	Right	5.0
Lip6	Right	17.0
Lip7	Rear	8.0
Lip8	Rear	14.0

MQ modules: two per board, PCB centres at X 9.0 and 13.0 (front board) and Y 28.85 and 32.85 (side boards), plugged into the breadboard row nearest the wall with the dome facing out. The dome projects about 1.2-1.4 cm past the breadboard edge, so it pushes through the 2 mm wall into the open air. Dry-fit one module first; the open-bottom port tolerates any row.

TIER 2 walls · Z from wall bottom (= Tier 2 floor) · ports open at the bottom so the box drops over the jutting sensors



Sheet 6 · Tier 3 compute deck

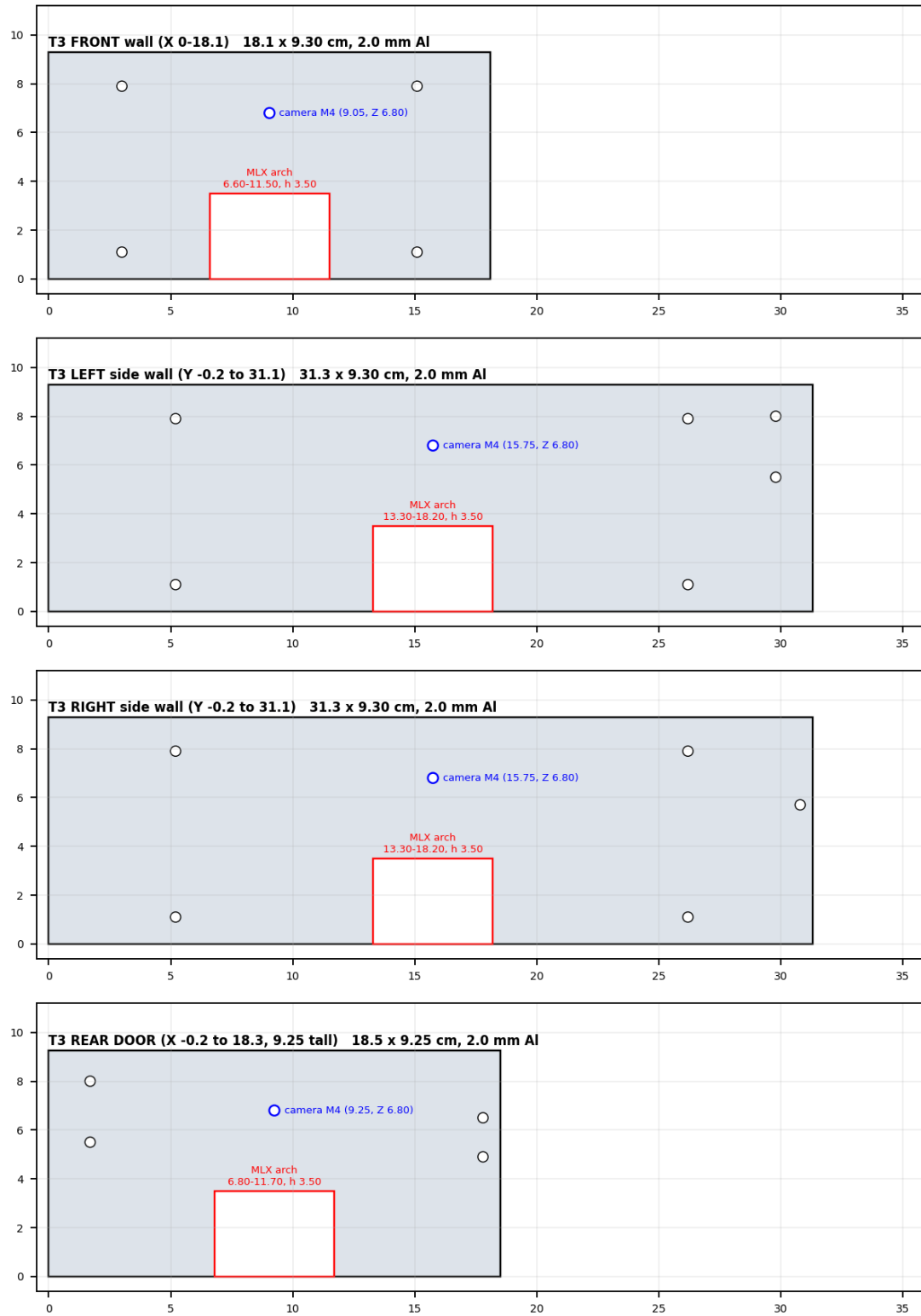


Part	X from-to	Y from-to	Height
Front mini + MLX90640	6.70 - 11.40	0.60 - 4.10	2.88
Rear mini + MLX90640	6.70 - 11.40	27.00 - 30.50	2.88
Left mini + MLX90640	0.40 - 3.90	13.20 - 17.90	2.88
Right mini + MLX90640	14.20 - 17.70	13.20 - 17.90	2.88
ESP32 breadboard	4.90 - 13.10	5.50 - 11.00	2.20
Raspberry Pi 5	4.80 - 13.30	12.00 - 17.60	3.00
USB hub bay (7-port)	2.20 - 14.00	19.00 - 26.50	3.00
Latch post	17.10 - 18.10	30.10 - 31.10	Z 4.5-9.0

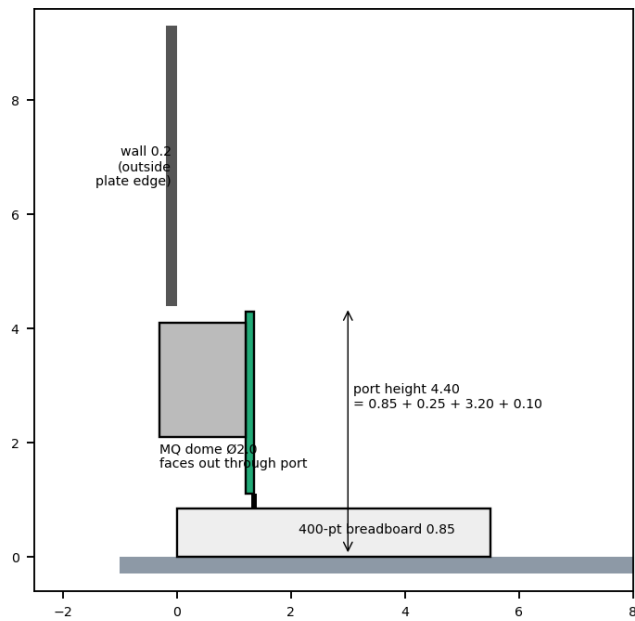
Bracket	Wall	Centre along wall (cm)
Lip1	Front	3.0
Lip2	Front	15.1
Lip3	Left	5.0

Bracket	Wall	Centre along wall (cm)
Lip4	Left	26.0
Lip5	Right	5.0
Lip6	Right	26.0
Lip7	Rear	3.0
Lip8	Rear	15.1

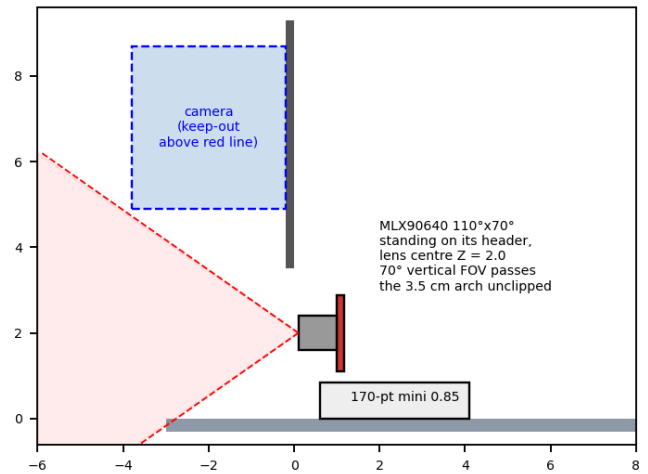
TIER 3 walls · Z from wall bottom (= Tier 3 floor) · arches 3.5 cm high, camera holes raised to Z 6.8



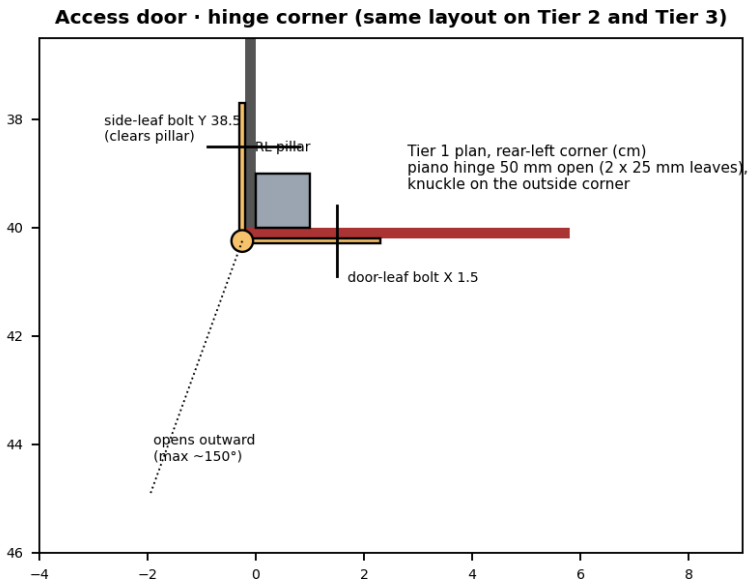
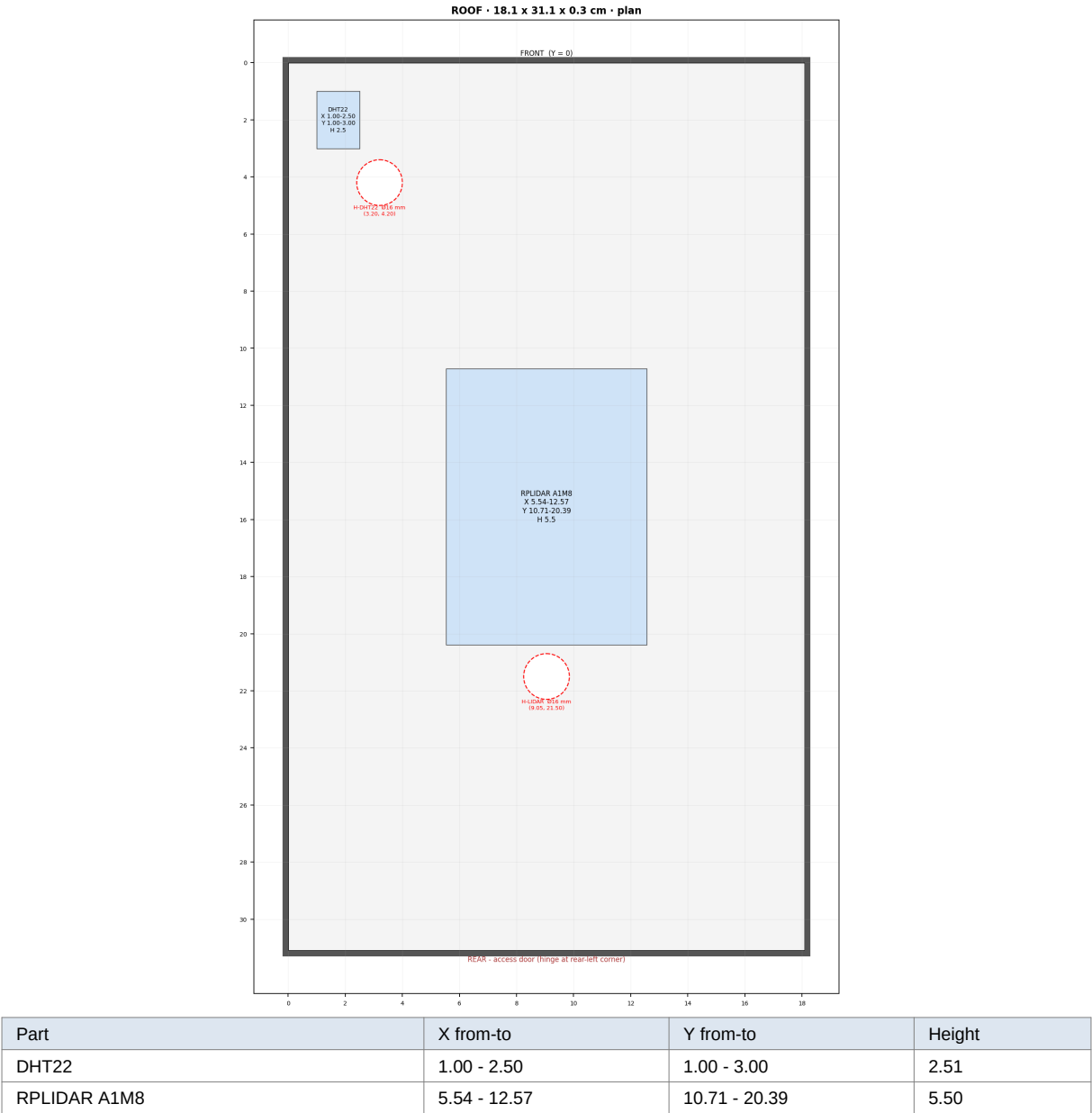
Tier 2 section · vertical MQ module at a port



Tier 3 section · thermal sensor at an arch



Sheet 7 · Roof and access doors



Door on every tier (rear wall). Piano hinge on the outside rear-left corner: side-wall leaf bolts at 1.7 cm from the rear edge (Tier 1: Y 38.5, in front of the pillar), door-leaf bolts 1.5 cm from the corner. Tier 1 hinge 8.0 cm long, bolts at Z 1.5 / 4.9 / 8.2. Tier 2/3 hinge 4.4 cm long, starting at Z 4.8 so it stays above the gas port; bolts at Z 5.5 / 8.0. Latch: 2 x M4x20 through the door into the tapped RR pillar (Tier 1, Z 3.9 / 6.9 on the door) or the latch post (Tier 2/3, Z 4.9 / 6.5). Limit opening to about 150° on Tier 2, because the MQ-8 dome juts out of the left wall near the hinge.

Sheet 8 · Assembly sequence

A · Machine everything flat

Cut all plates and walls (sheet 2). Drill every Ø4.2 hole from the drawings; tap pillar ends 20 mm deep; drill pillar and post through-holes; mill the gas ports and thermal arches; drill camera holes at Z 6.8. Deburr every edge (12 V bus safety).

B · Tier 1 skeleton

1. Bolt the 4 pillars to the base plate from underneath (M4x20 into the pillar threads, Loctite blue). 2. Fit B1-B12 on the base plate, horizontal leg inward, outer face of the vertical leg flush with the plate edge; full stack with rubber washer. 3. Fit the Tier 1 components (sheet 4), Battery C bay and regulator bay. 4. Clamp the side walls and front wall against the brackets and pillars; bolt wall-to-bracket and wall-to-pillar. 5. Fit U1-U12 to the walls at the top, horizontal leg level with the pillar tops. 6. Lay the mid plate on the pillars and U-brackets; bolt into the pillars (Loctite) and the brackets (nyloc underneath), cross pattern FL-RR-FR-RL. 7. Hang the rear door on its hinge; leave it OPEN.

C · Tier 2

1. Stick the four breadboards to the mid plate at sheet 5 coordinates; plug in the Nano and the 6 MQ modules (vertical, domes to the walls); wire per Part 2 sheet A; run the Tier-2 trunk up through H-T2. 2. Build the box upside down: roof brackets RB1-RB8 to the Tier 2 roof, then the front and side walls, then the latch post. 3. Lower the box over the circuitry: the open-bottom ports slide past the jutting domes. 4. Fit lips Lip1-Lip8 inside, reaching in through the open rear; bolt each lip to the wall, then down through the mid plate (nyloc from inside Tier 1 through the Tier 1 door). 5. Hang the Tier 2 door; leave it open.

D · Tier 3

1. On the Tier 2 roof: Pi 5 on its standoffs, ESP32 board, the 4 minis with MLX90640s standing on their headers, lenses facing the arch centres, and the USB hub. Run Trunk 1, Trunk 2, the Tier 1 and Tier 2 Nano USB leads up through H-T3. 2. Build the Tier 3 box the same way (roof brackets, walls, latch post); fit the cameras on their 15° brackets at the Z 6.8 holes. 3. Lower it on; check each lens looks straight out of its arch. 4. Bolt the lips (nuts inside Tier 2 through the Tier 2 door).

E · Roof and close-up

1. Fit the RPLIDAR at dead centre (match-drill from the unit) and the DHT22 at front-left; drop their leads through H-LIDAR and H-DHT22. 2. Close the doors top-down: Tier 3, Tier 2, then Tier 1. Fit 2 latch bolts each. 3. Power-on order: 12 V and Battery B first (or together with Battery C).

Wire routes (electrical design unchanged from Part 1 / Part 2)

Run	Path
12 V	LiPo -> 12 V bus -> both drivers B+ ; GND bus -> drivers B-. Motor leads go down through base-plate grommets (drivetrain document)
Battery B 5 V	Battery B -> 5 V bus -> Tier 1 rails. Tier-2 trunk: 5 V bus -> H-T2 -> centre Nano board rails. Trunk 1: 5 V bus -> H-T3 -> H-T3 (Tier 2 roof) -> ESP32 board
Battery C	Pack -> 5.1 V regulator (right of the 12 V bus) -> Trunk 2 (18 AWG, keep under 50 cm) -> H-T3 -> H-T3 -> Pi 5 USB-C (stripped official cable) + hub DC terminal. One GND link to the master GND bus
USB	Tier 1 Nano -> H-T3 -> Pi USB 3.0 #2; Tier 2 Nano -> H-T3 (Tier 2 roof) -> Pi USB 2.0 #2; ESP32 -> Pi USB 2.0 #1; hub -> Pi USB 3.0 #1; 4 cameras + LIDAR adapter -> hub
Roof	DHT22 3 wires -> H-DHT22 -> ESP32 3V3/GND rails + GPIO 4; LIDAR cable -> H-LIDAR -> USB adapter -> hub

Sheet 9 · Verification and checks before cutting metal

Script checks over the single coordinate model: every part inside its deck; no two parts overlap in 3D; no pass-through hole under a part; no lip bolt tip hits a part in the deck below; no bracket inside a sensor port; thermal field of view through the arches; M4x20 length at every joint type.

Result
MLX FOV: needs 2.90 cm high x 3.26 cm wide, arch 3.5 x 4.9 -> OK
Camera hole Z=6.80 cm: with a 3.8 cm tall camera, bottom at 4.90 cm clears the MLX FOV for up to 3.9 cm protrusion
Stack Plate 0.3 + bracket 0.2 protrusion past nyloc 0.82 cm OK
Stack Base 0.4 + bracket 0.2 protrusion past nyloc 0.72 cm OK
Stack Wall 0.2 + bracket 0.2 protrusion past nyloc 0.92 cm OK
Stack Wall 0.2 + pillar 1.0 (through) protrusion past nyloc 0.14 cm OK
Stack Hinge leaf 0.1 + wall 0.2 protrusion past nyloc 1.04 cm OK
Pillar tap engagement, base 0.4: 1.52 cm (tap depth 2.0) -> OK
TOTAL ERRORS: 0

Things you must confirm on the real parts

#	Check
1	USB-HUB-7U outline fits the 11.8 x 7.5 x 3.0 cm bay (Waveshare gives it only as a drawing image).
2	Battery C pack case $\leq 13.0 \times 7.0 \times 4.5$ cm and the regulator $\leq 6.0 \times 4.5 \times 2.5$ cm; regulator rated ≥ 8 A at 5.1 V.
3	Your MQ header lets the module stand vertical (straight in-plane or 90° header). Dry-fit one module at a port.
4	MLX90640 lens centre sits about 2.0 cm above the deck when standing on its header; the arch allows 2.9 cm at worst case.
5	Camera body: its underside must stay above the red line on the section drawing (at most 3.9 cm protrusion with a 3.8 cm tall camera at Z 6.8).
6	BTS7960 and bus-bar hole patterns: match-drill from your actual boards; their footprints are as in your layout.